Pikachu Metrics 数据分析 metrics_Mjlab-Tracking-Flat-Pikachu-Up__model_4999_20260820_102758

FILE
metrics_Mjlab-Tracking-Flat-Pikachu-Up__model_4999_20260820_102758.csv

NETWORK
Mjlab-Tracking-Flat-Pikachu-Up__model_4999

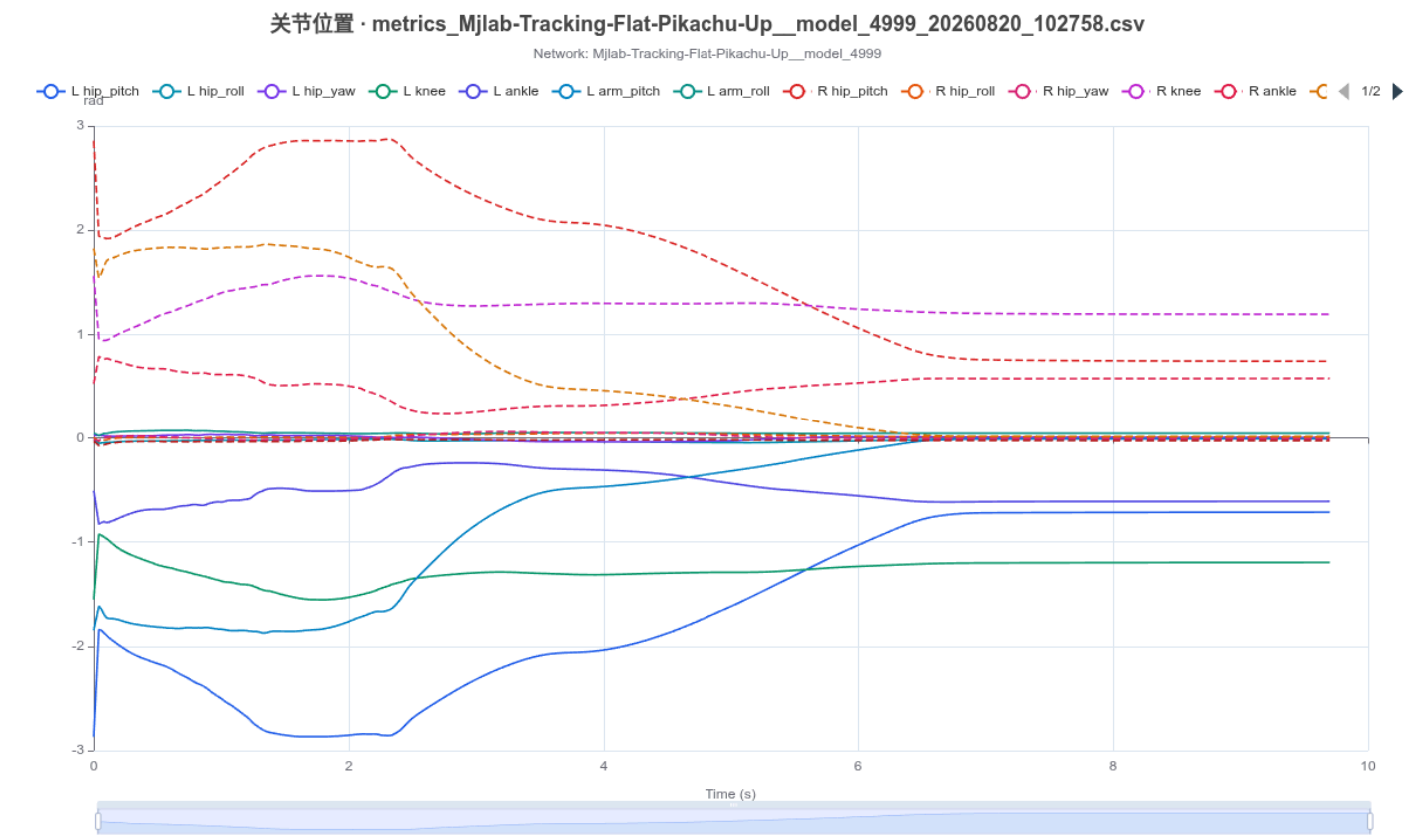
SAMPLES
455

DURATION
9.700 s

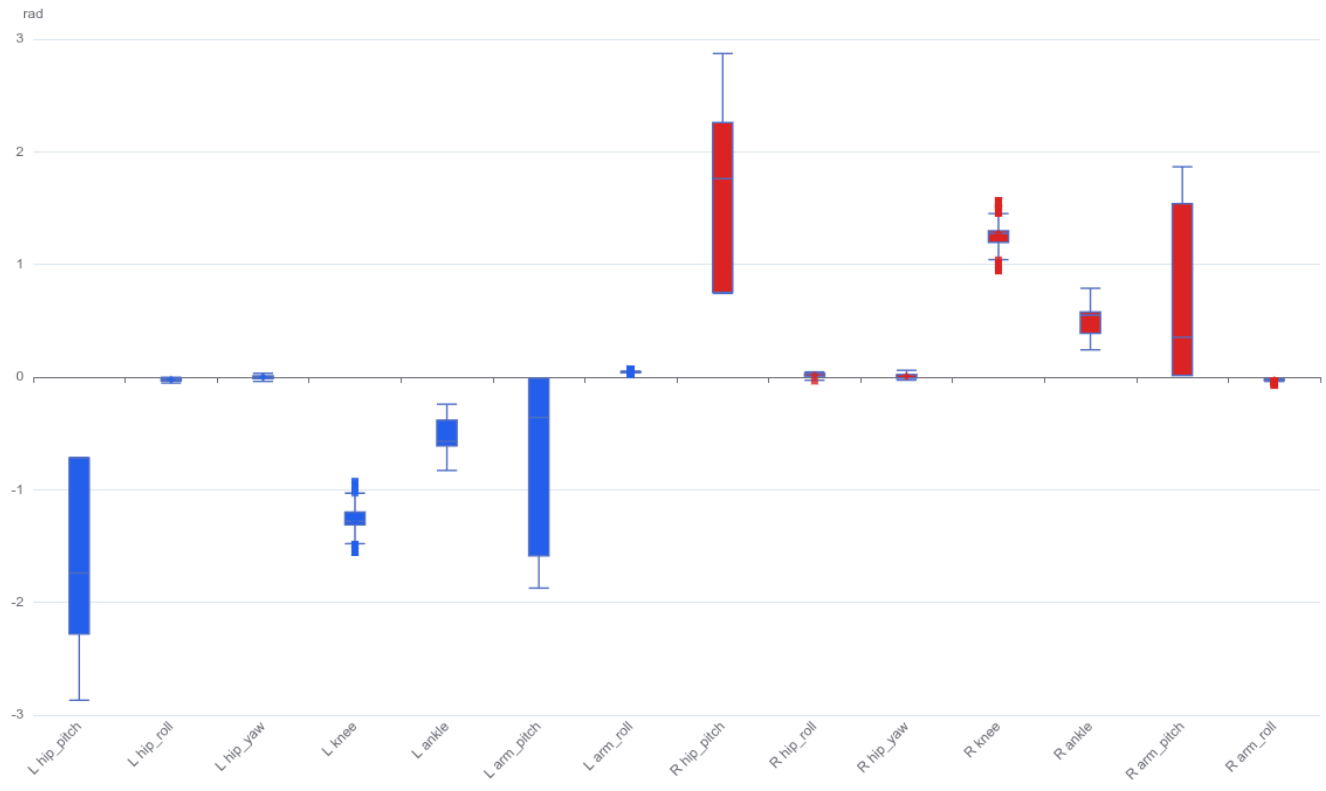
ENVIRONMENTS
1

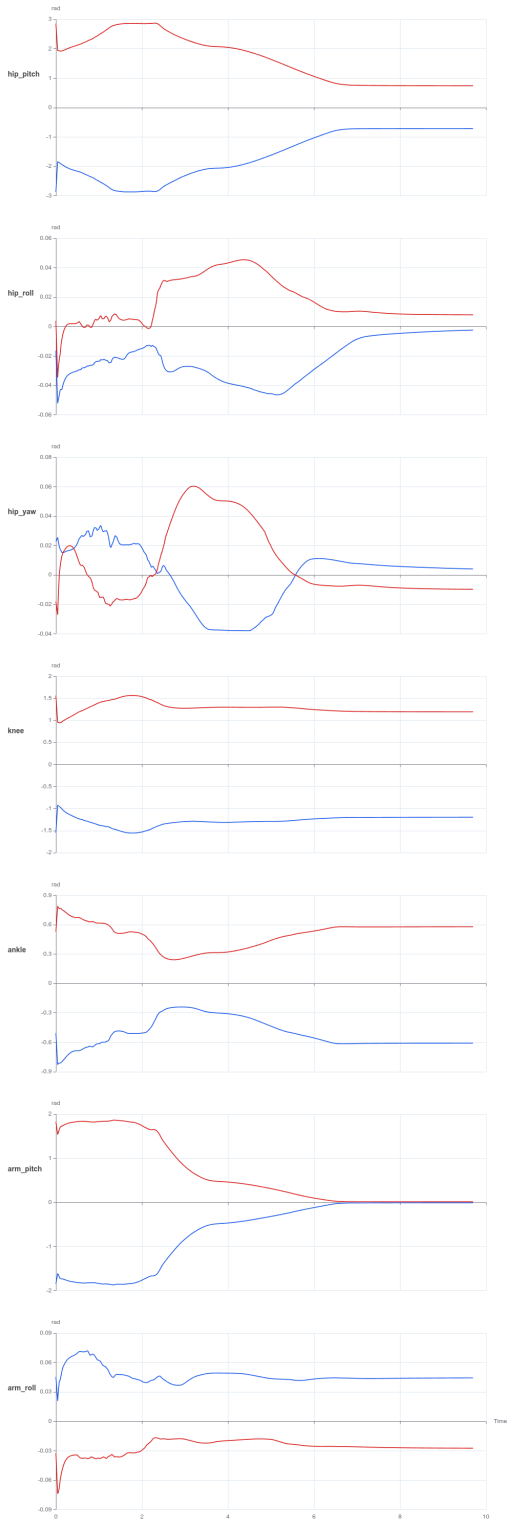
时间区间: 0.000 — 9.700 s · 仅含已勾选字段

关节位置

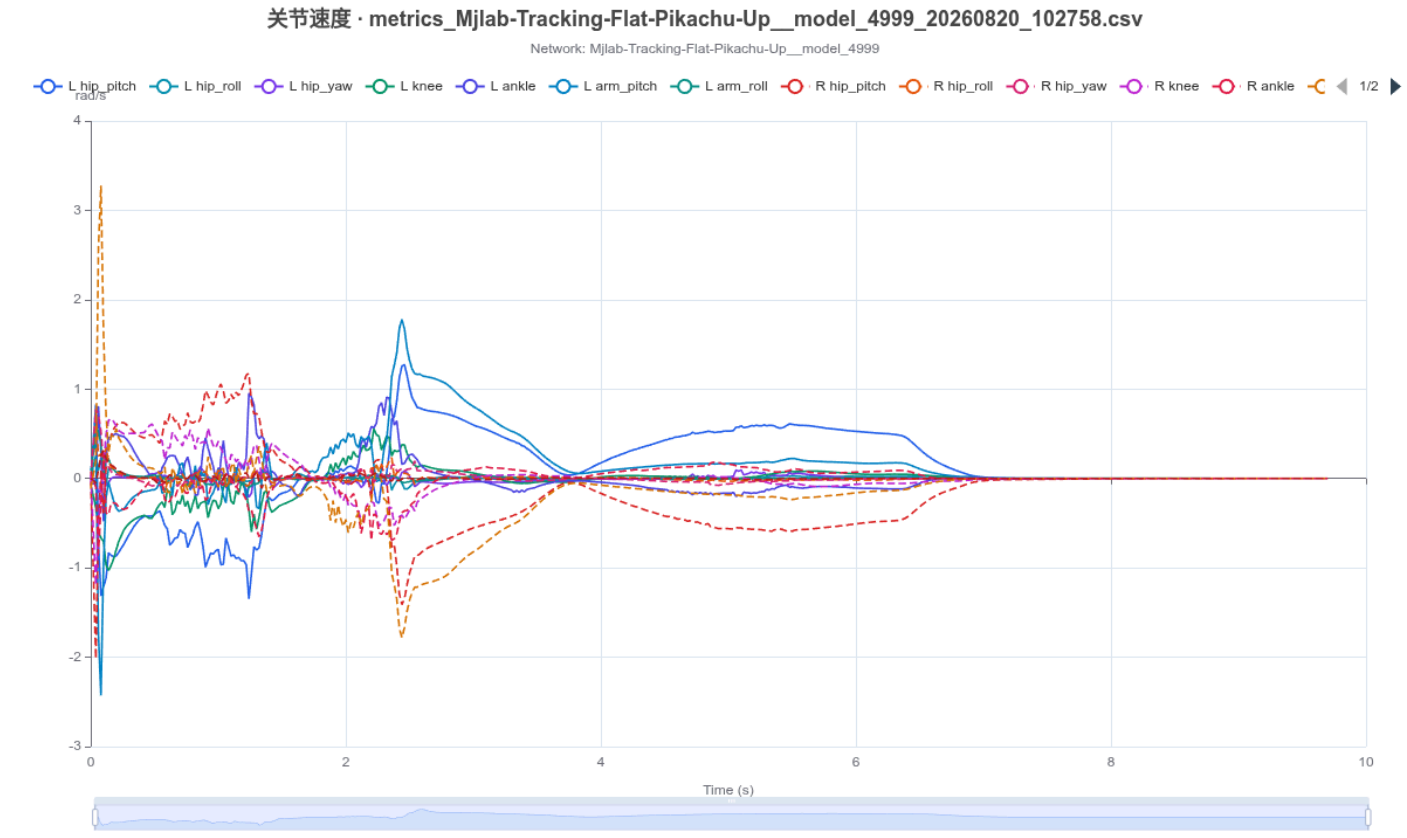


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	455	-2.86828	-0.713651	-1.62805	1.81366	2.86828	2.85559
joint_position_rad/left_hip_roll_joint	10	15	455	-0.0521939	-0.00241651	-0.0224378	0.0266419	0.0521939	0.0453010
joint_position_rad/left_hip_yaw_joint	10	15	455	-0.0379674	0.0335969	-0.000169570	0.0202547	0.0379674	0.0377789
joint_position_rad/left_knee_joint	30	10	455	-1.55377	-0.923590	-1.27817	1.28269	1.55377	1.52594
joint_position_rad/left_ankle_joint	10	15	455	-0.828365	-0.240346	-0.508417	0.528123	0.828365	0.685599
joint_position_rad/left_arm_pitch_joint	10	15	455	-1.87313	-0.00964809	-0.654933	0.978017	1.87313	1.84947
joint_position_rad/left_arm_roll_joint	10	15	455	0.0208578	0.0718691	0.0465732	0.0471504	0.0718691	0.0669116
joint_position_rad/right_hip_pitch_joint	30	10	455	0.742850	2.87240	1.64324	1.82022	2.87240	2.85777
joint_position_rad/right_hip_roll_joint	10	15	455	-0.0346201	0.0454276	0.0170903	0.0228406	0.0454276	0.0441443
joint_position_rad/right_hip_yaw_joint	10	15	455	-0.0270774	0.0603235	0.00795758	0.0257211	0.0603235	0.0544081
joint_position_rad/right_knee_joint	30	10	455	0.942593	1.56226	1.27531	1.28041	1.56226	1.53439
joint_position_rad/right_ankle_joint	10	15	455	0.241476	0.788344	0.497466	0.513399	0.788344	0.670712
joint_position_rad/right_arm_pitch_joint	10	15	455	0.0133102	1.86669	0.649486	0.972770	1.86669	1.83902
joint_position_rad/right_arm_roll_joint	10	15	455	-0.0737150	-0.0165285	-0.0262636	0.0273015	0.0737150	0.0376626

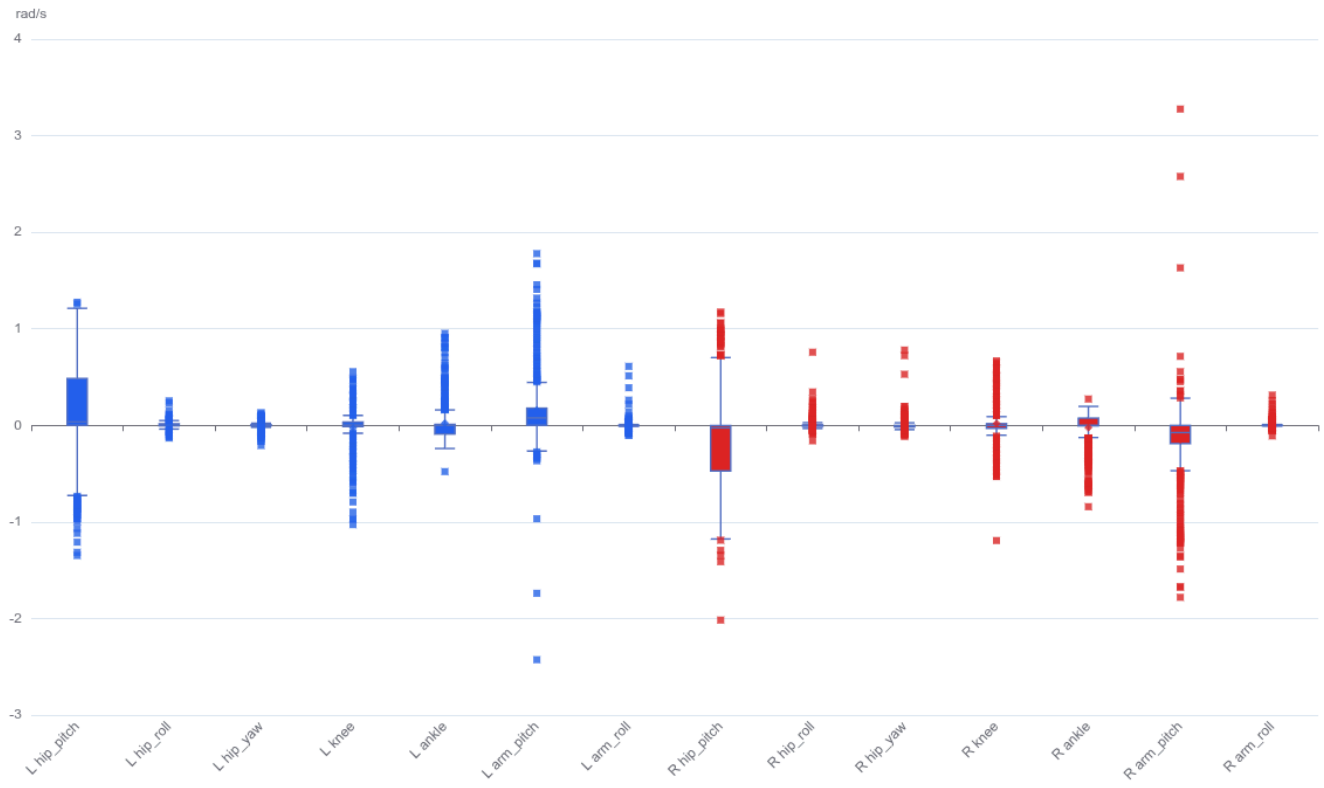

rad
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

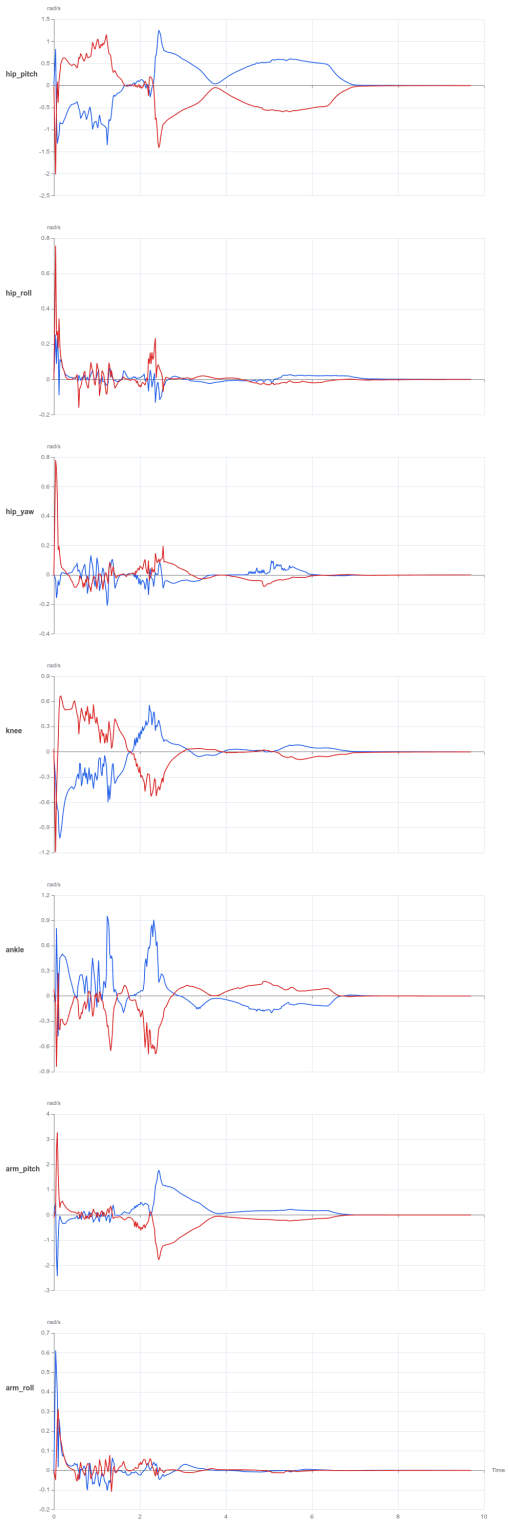

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

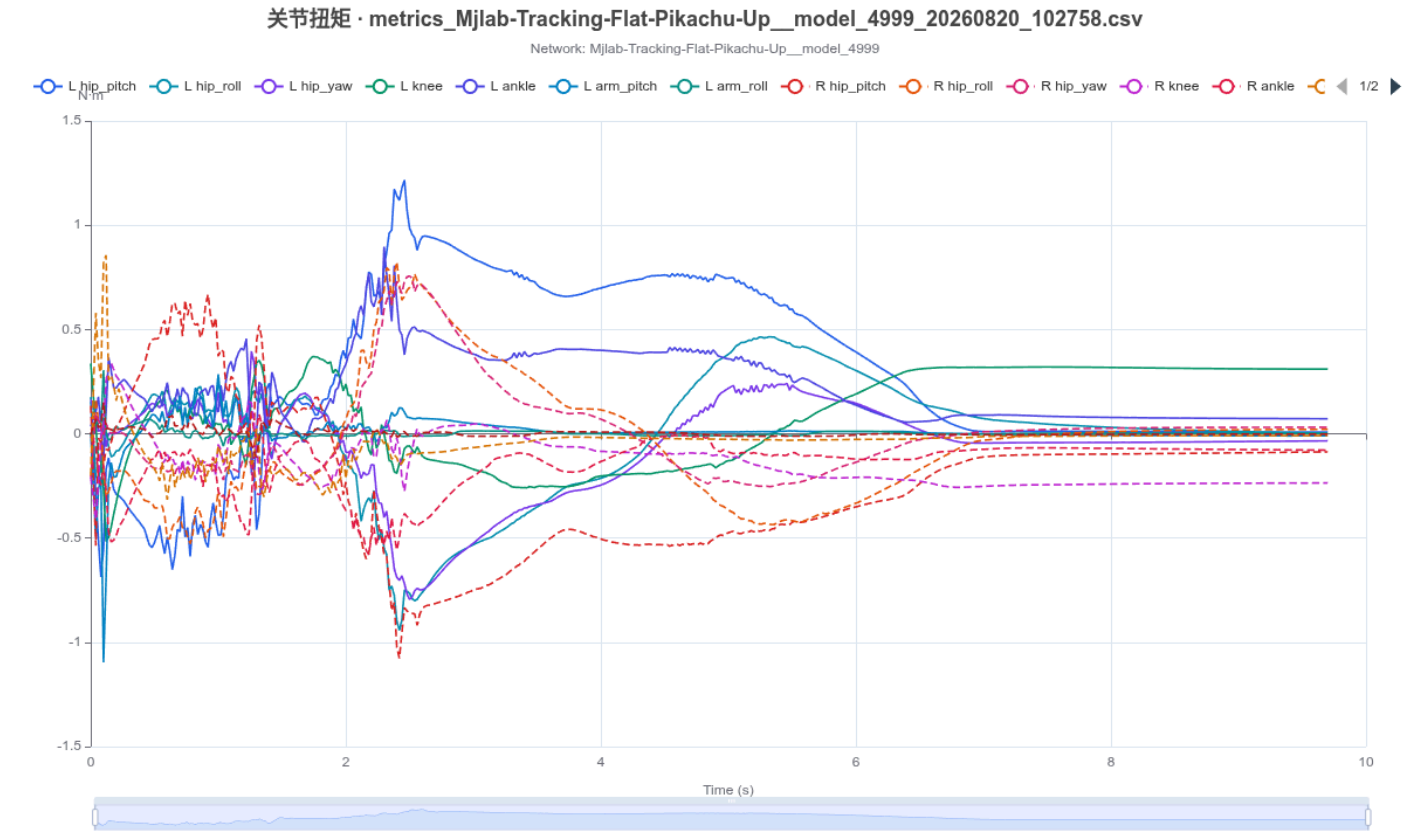


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	455	-1.34755	1.27531	0.107983	0.453209	1.34755	0.866467
joint_velocity_rad_s/left_hip_roll_joint	10	15	455	-0.131154	0.254298	0.00542691	0.0288197	0.254298	0.0494884
joint_velocity_rad_s/left_hip_yaw_joint	10	15	455	-0.208667	0.132081	-0.00162924	0.0373052	0.208667	0.0806425
joint_velocity_rad_s/left_knee_joint	30	10	455	-1.02828	0.556437	-0.0270570	0.195597	1.02828	0.443343
joint_velocity_rad_s/left_ankle_joint	10	15	455	-0.478527	0.953019	0.0237575	0.194965	0.953019	0.467431
joint_velocity_rad_s/left_arm_pitch_joint	10	15	455	-2.42657	1.77879	0.156799	0.394342	2.42657	1.06556
joint_velocity_rad_s/left_arm_roll_joint	10	15	455	-0.102710	0.611372	0.00306396	0.0493688	0.611372	0.0598655
joint_velocity_rad_s/right_hip_pitch_joint	30	10	455	-2.01621	1.17395	-0.119905	0.448671	2.01621	0.868362
joint_velocity_rad_s/right_hip_roll_joint	10	15	455	-0.159794	0.757669	0.00610000	0.0545095	0.757669	0.0836007
joint_velocity_rad_s/right_hip_yaw_joint	10	15	455	-0.114880	0.780129	0.00318044	0.0679602	0.780129	0.0875817
joint_velocity_rad_s/right_knee_joint	30	10	455	-1.19242	0.668090	0.0195569	0.201938	1.19242	0.502557
joint_velocity_rad_s/right_ankle_joint	10	15	455	-0.842077	0.273011	-0.0231050	0.165941	0.842077	0.380710
joint_velocity_rad_s/right_arm_pitch_joint	10	15	455	-1.78055	3.27601	-0.147976	0.426603	3.27601	1.09420
joint_velocity_rad_s/right_arm_roll_joint	10	15	455	-0.109194	0.313663	0.00450574	0.0304828	0.313663	0.0487537

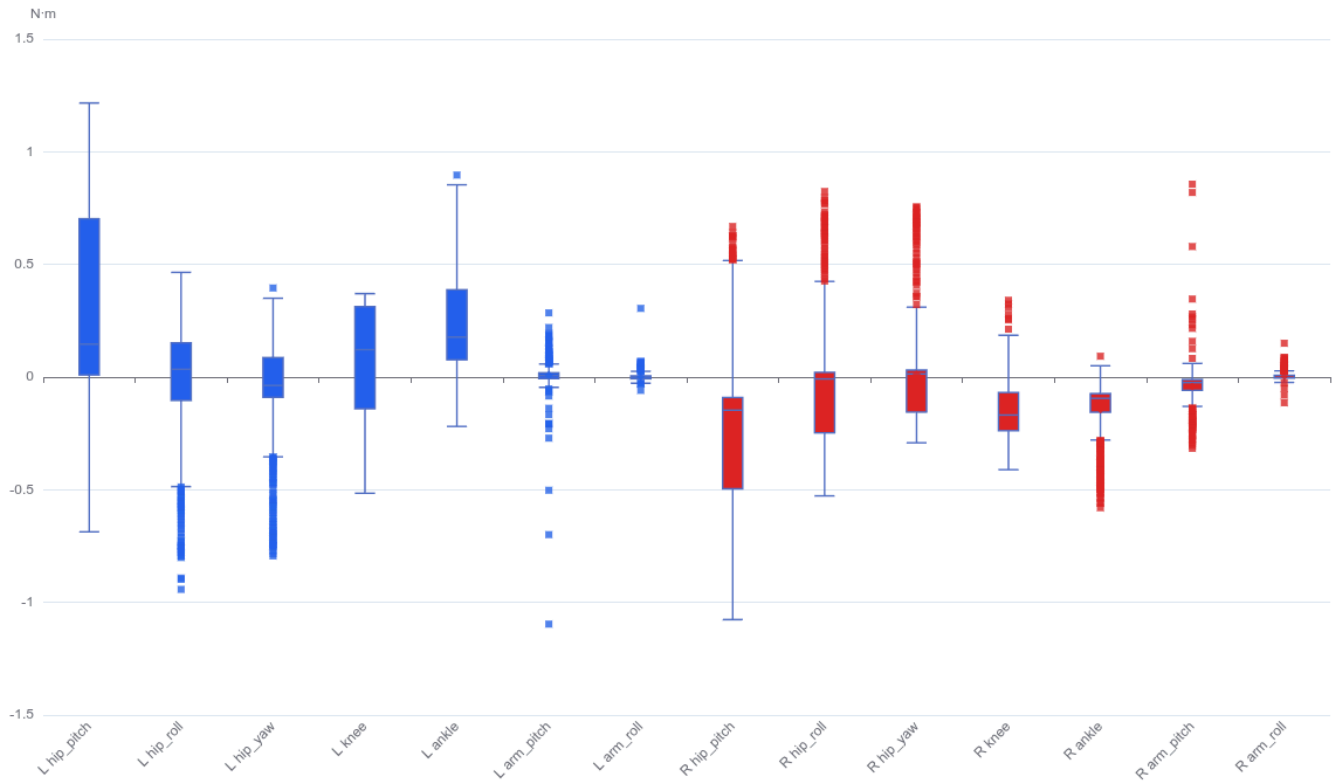

rad/s
4
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

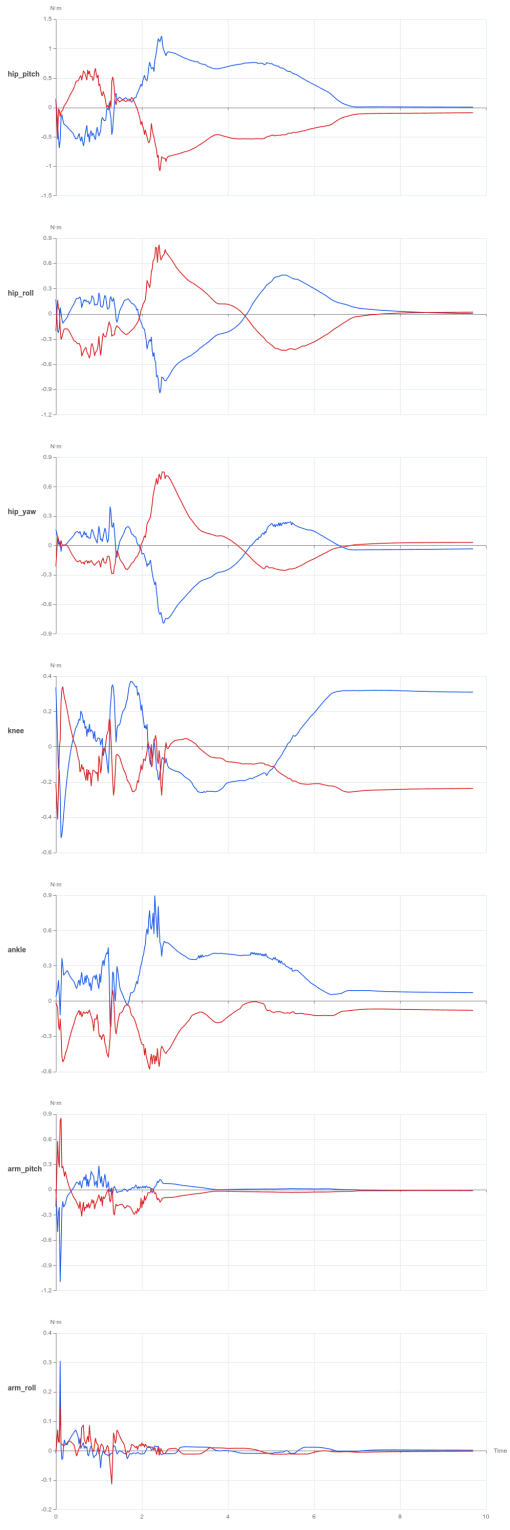

rad/s
hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节扭矩



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	455	-0.686619	1.21645	0.280911	0.509494	1.21645	0.905187
joint_torque_nm/left_hip_roll_joint	10	15	455	-0.942198	0.464843	-0.00785637	0.286515	0.942198	0.606043
joint_torque_nm/left_hip_yaw_joint	10	15	455	-0.793721	0.395475	-0.0624139	0.237681	0.793721	0.602103
joint_torque_nm/left_knee_joint	30	10	455	-0.515692	0.370397	0.0883850	0.239841	0.515692	0.320193
joint_torque_nm/left_ankle_joint	10	15	455	-0.219153	0.896666	0.226921	0.284448	0.896666	0.487587
joint_torque_nm/left_arm_pitch_joint	10	15	455	-1.09625	0.285024	0.00905229	0.0828063	1.09625	0.124492
joint_torque_nm/left_arm_roll_joint	10	15	455	-0.0589602	0.304987	0.00242567	0.0194662	0.304987	0.0258928
joint_torque_nm/right_hip_pitch_joint	30	10	455	-1.07635	0.668456	-0.233208	0.413982	1.07635	0.791320
joint_torque_nm/right_hip_roll_joint	10	15	455	-0.527404	0.823914	-0.0347333	0.290217	0.823914	0.610185
joint_torque_nm/right_hip_yaw_joint	10	15	455	-0.291544	0.756312	0.0123042	0.219759	0.756312	0.587770
joint_torque_nm/right_knee_joint	30	10	455	-0.411234	0.341035	-0.137885	0.179753	0.411234	0.253316
joint_torque_nm/right_ankle_joint	10	15	455	-0.579769	0.0927118	-0.144383	0.191291	0.579769	0.448966
joint_torque_nm/right_arm_pitch_joint	10	15	455	-0.316267	0.856140	-0.0421173	0.109580	0.856140	0.221092
joint_torque_nm/right_arm_roll_joint	10	15	455	-0.113852	0.150269	0.00287434	0.0185545	0.150269	0.0326531

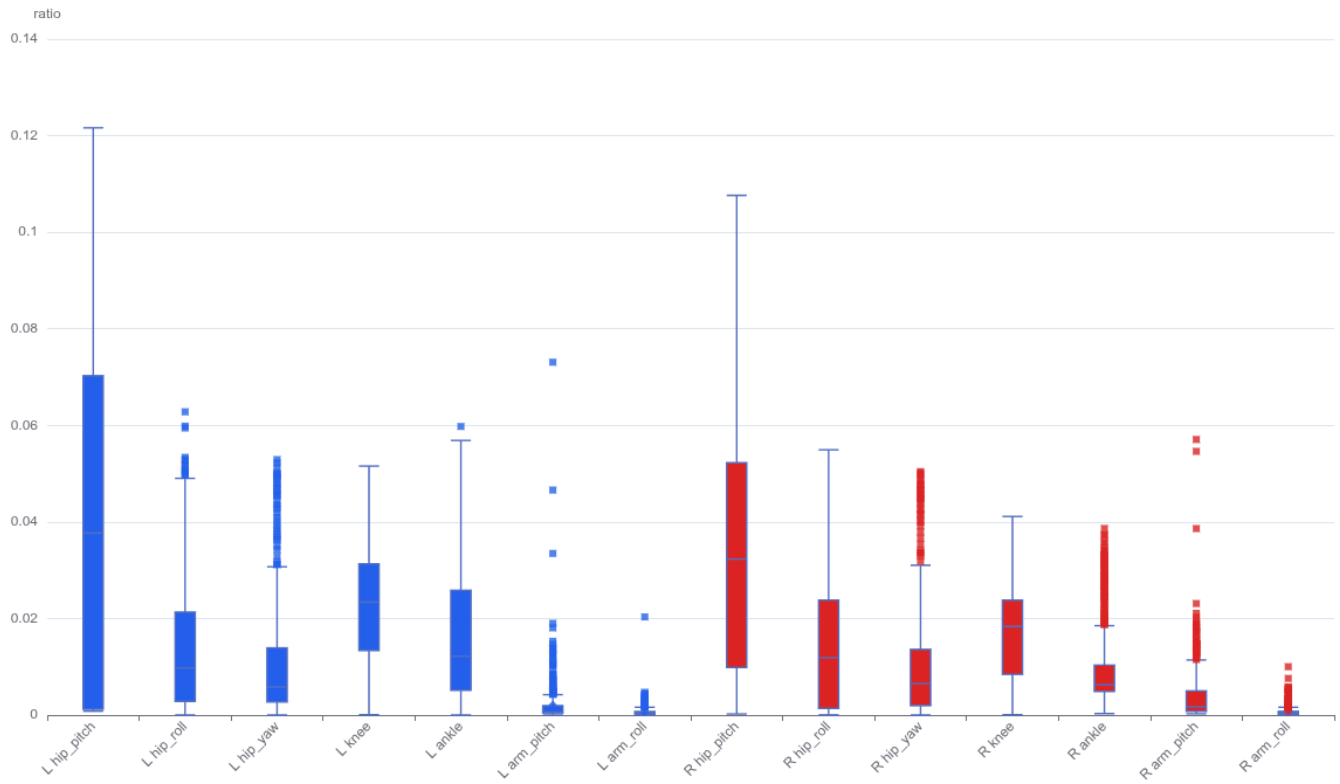

N·m
1.5
1
0.5
0
-0.5
-1
-1.5
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

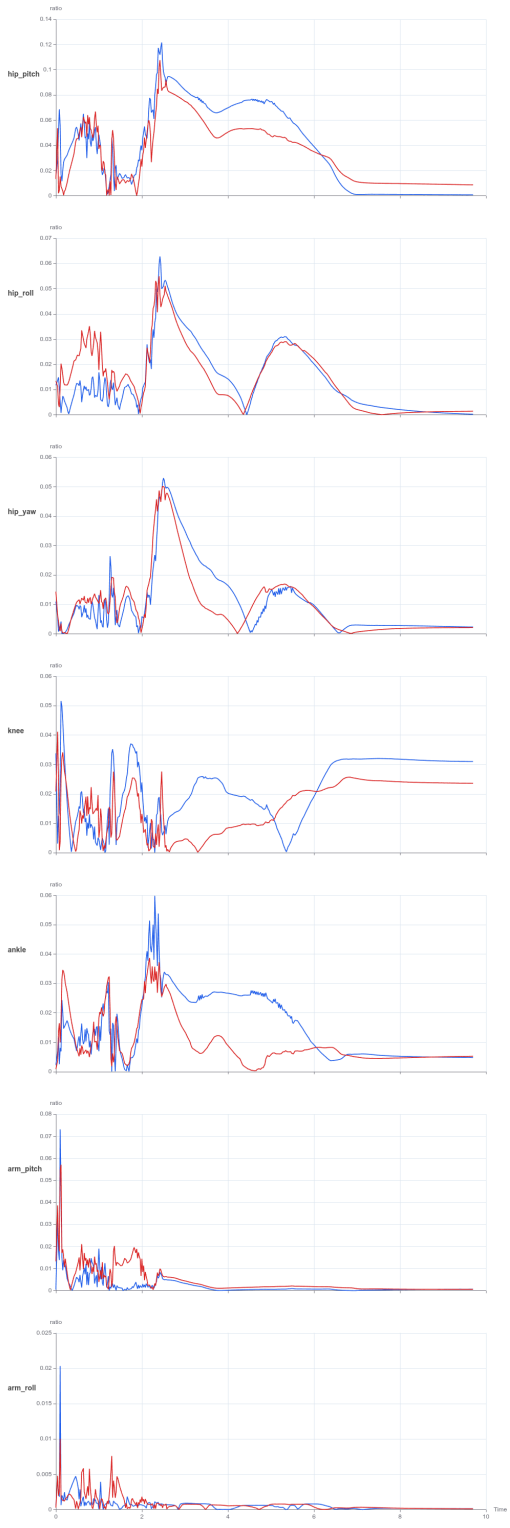

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节扭矩占用

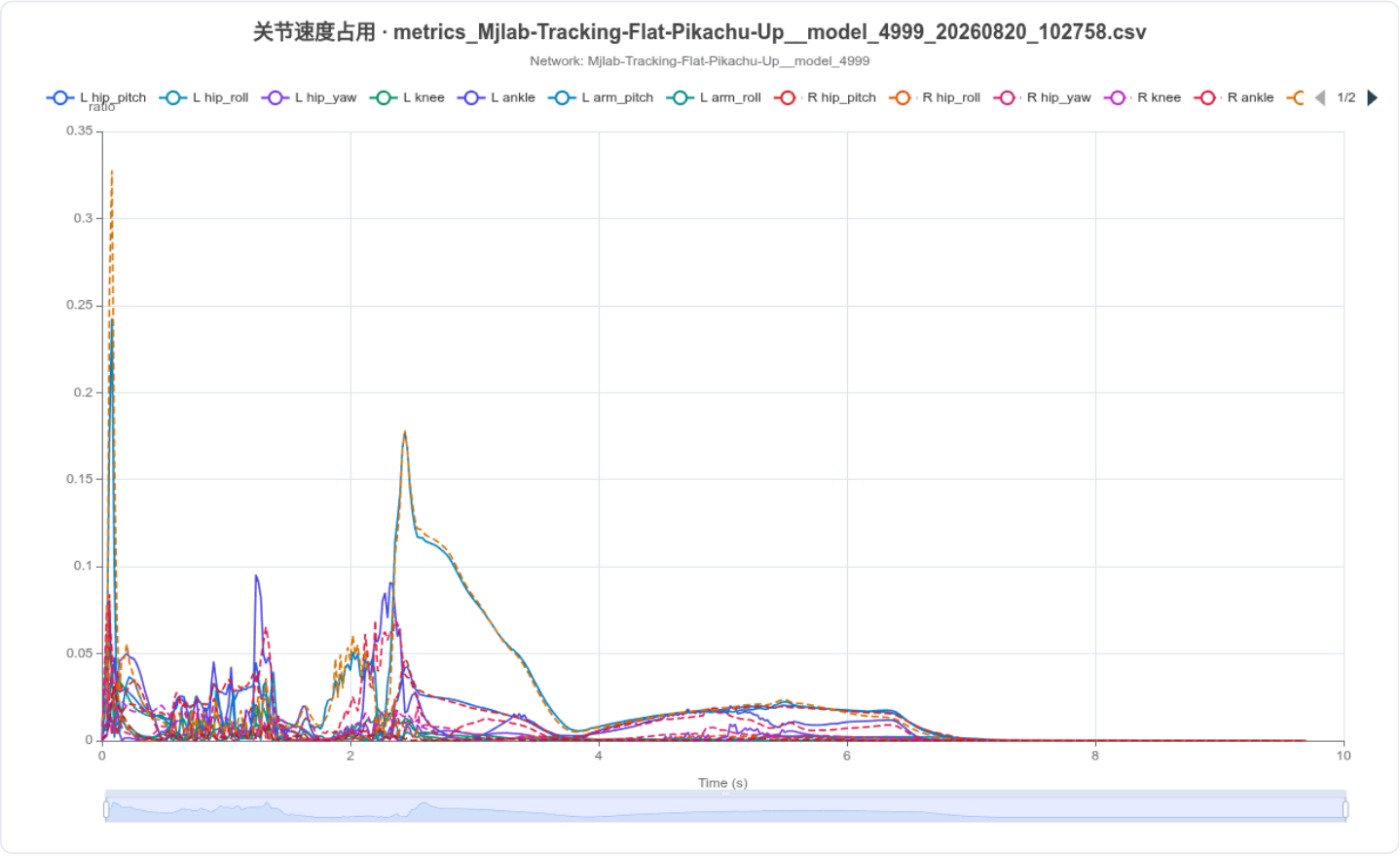


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	455	0.000798634	0.121645	0.0385984	0.0509494	0.121645	0.0905187
joint_torque_ratio/left_hip_roll_joint	10	15	455	0.0000243013	0.0628132	0.0136824	0.0191010	0.0628132	0.0404029
joint_torque_ratio/left_hip_yaw_joint	10	15	455	0.0000236970	0.0529148	0.0106368	0.0158454	0.0529148	0.0401402
joint_torque_ratio/left_knee_joint	30	10	455	0.0000599057	0.0515692	0.0216238	0.0239841	0.0515692	0.0320194
joint_torque_ratio/left_ankle_joint	10	15	455	0.0000260349	0.0597778	0.0153089	0.0189632	0.0597778	0.0325058
joint_torque_ratio/left_arm_pitch_joint	10	15	455	7.50381e-7	0.0730835	0.00215467	0.00552042	0.0730835	0.00829948
joint_torque_ratio/left_arm_roll_joint	10	15	455	3.47847e-7	0.0203325	0.000610987	0.00129775	0.0203325	0.00172619
joint_torque_ratio/right_hip_pitch_joint	30	10	455	0.000238187	0.107635	0.0337458	0.0413982	0.107635	0.0791320
joint_torque_ratio/right_hip_roll_joint	10	15	455	0.0000293416	0.0549276	0.0143069	0.0193478	0.0549276	0.0406790
joint_torque_ratio/right_hip_yaw_joint	10	15	455	5.11457e-8	0.0504208	0.00970443	0.0146506	0.0504208	0.0391847
joint_torque_ratio/right_knee_joint	30	10	455	0.0000806938	0.0411234	0.0157969	0.0179753	0.0411234	0.0253316
joint_torque_ratio/right_ankle_joint	10	15	455	0.000304318	0.0386513	0.00967459	0.0127527	0.0386513	0.0299310
joint_torque_ratio/right_arm_pitch_joint	10	15	455	0.000263092	0.0570760	0.00409013	0.00730533	0.0570760	0.0147394
joint_torque_ratio/right_arm_roll_joint	10	15	455	0.00000332067	0.0100179	0.000710089	0.00123697	0.0100179	0.00217687

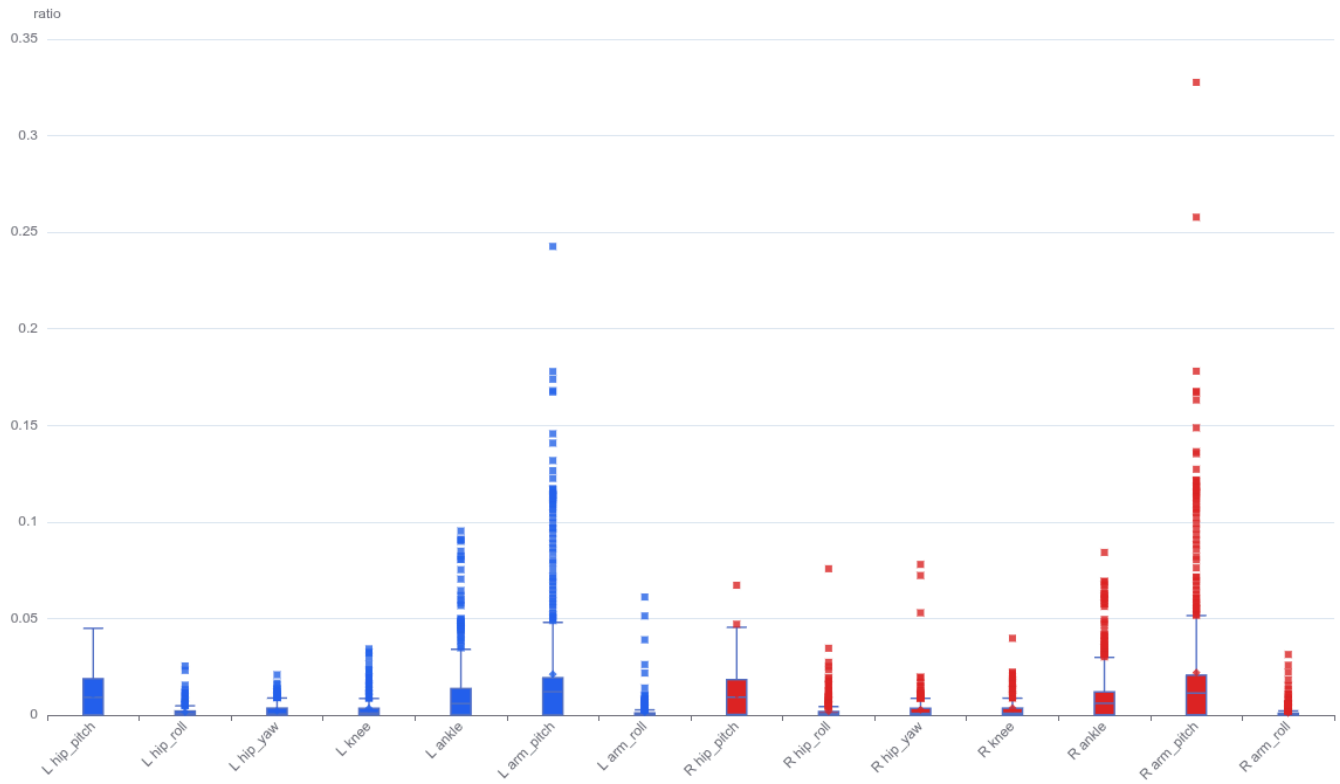

ratio
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

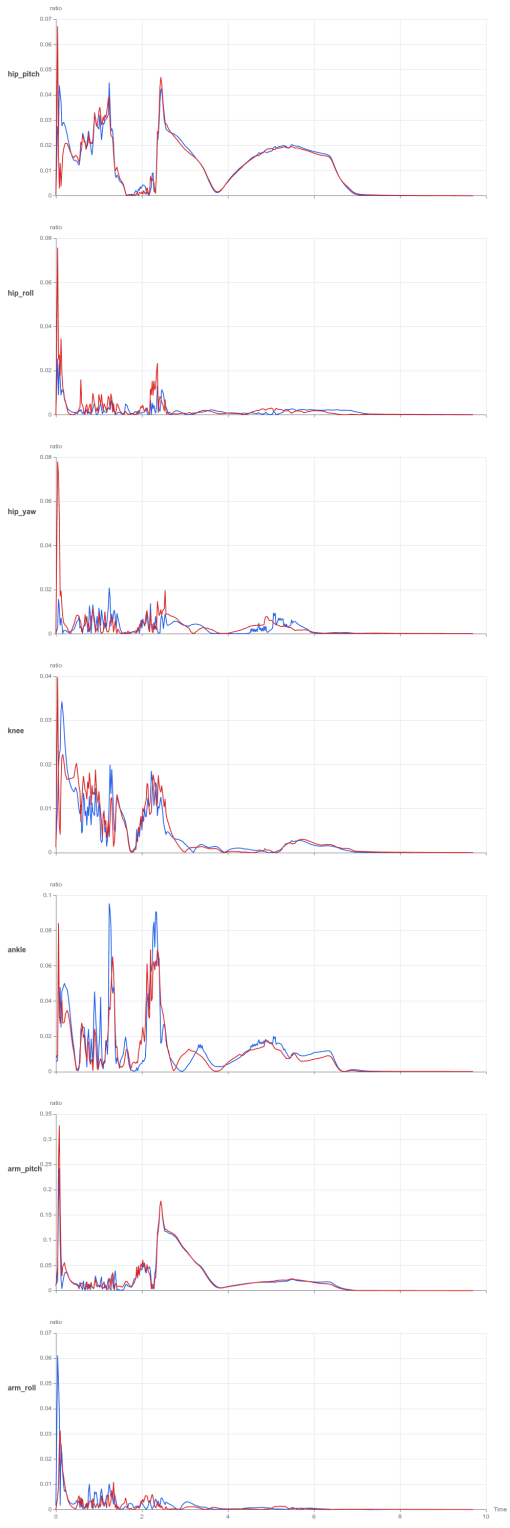

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用

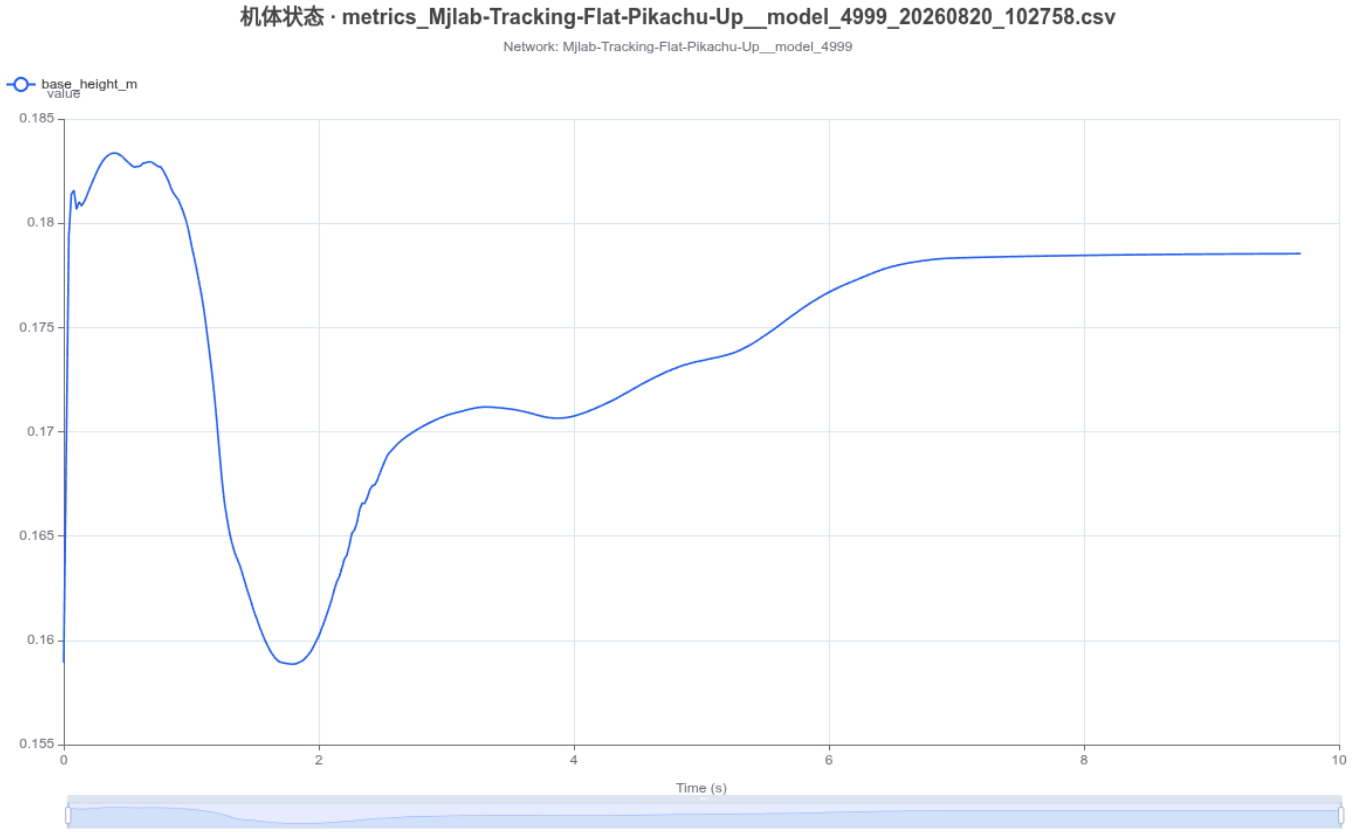


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	455	0.0000163919	0.0449185	0.0108455	0.0151070	0.0449185	0.0288822
joint_velocity_ratio/left_hip_roll_joint	10	15	455	6.45961e-7	0.0254298	0.00152212	0.00288197	0.0254298	0.00494884
joint_velocity_ratio/left_hip_yaw_joint	10	15	455	0.00000525434	0.0208667	0.00217018	0.00373052	0.0208667	0.00806425
joint_velocity_ratio/left_knee_joint	30	10	455	0.00000839727	0.0342759	0.00343542	0.00651990	0.0342759	0.0147781
joint_velocity_ratio/left_ankle_joint	10	15	455	0.0000247692	0.0953019	0.0108476	0.0194965	0.0953019	0.0467431
joint_velocity_ratio/left_arm_pitch_joint	10	15	455	0.0000111177	0.242657	0.0211700	0.0394342	0.242657	0.106556
joint_velocity_ratio/left_arm_roll_joint	10	15	455	0.00000308328	0.0611372	0.00148175	0.00493688	0.0611372	0.00598655
joint_velocity_ratio/right_hip_pitch_joint	30	10	455	0.0000231786	0.0672070	0.0106282	0.0149557	0.0672070	0.0289454
joint_velocity_ratio/right_hip_roll_joint	10	15	455	0.00000370163	0.0757669	0.00201525	0.00545095	0.0757669	0.00836007
joint_velocity_ratio/right_hip_yaw_joint	10	15	455	0.00000847293	0.0780129	0.00274316	0.00679602	0.0780129	0.00875817
joint_velocity_ratio/right_knee_joint	30	10	455	7.78313e-7	0.0397474	0.00360939	0.00673126	0.0397474	0.0167519
joint_velocity_ratio/right_ankle_joint	10	15	455	3.21867e-7	0.0842077	0.00956739	0.0165941	0.0842077	0.0380710
joint_velocity_ratio/right_arm_pitch_joint	10	15	455	7.26064e-7	0.327601	0.0219131	0.0426603	0.327601	0.109420
joint_velocity_ratio/right_arm_roll_joint	10	15	455	4.78569e-7	0.0313663	0.00109727	0.00304828	0.0313663	0.00487537

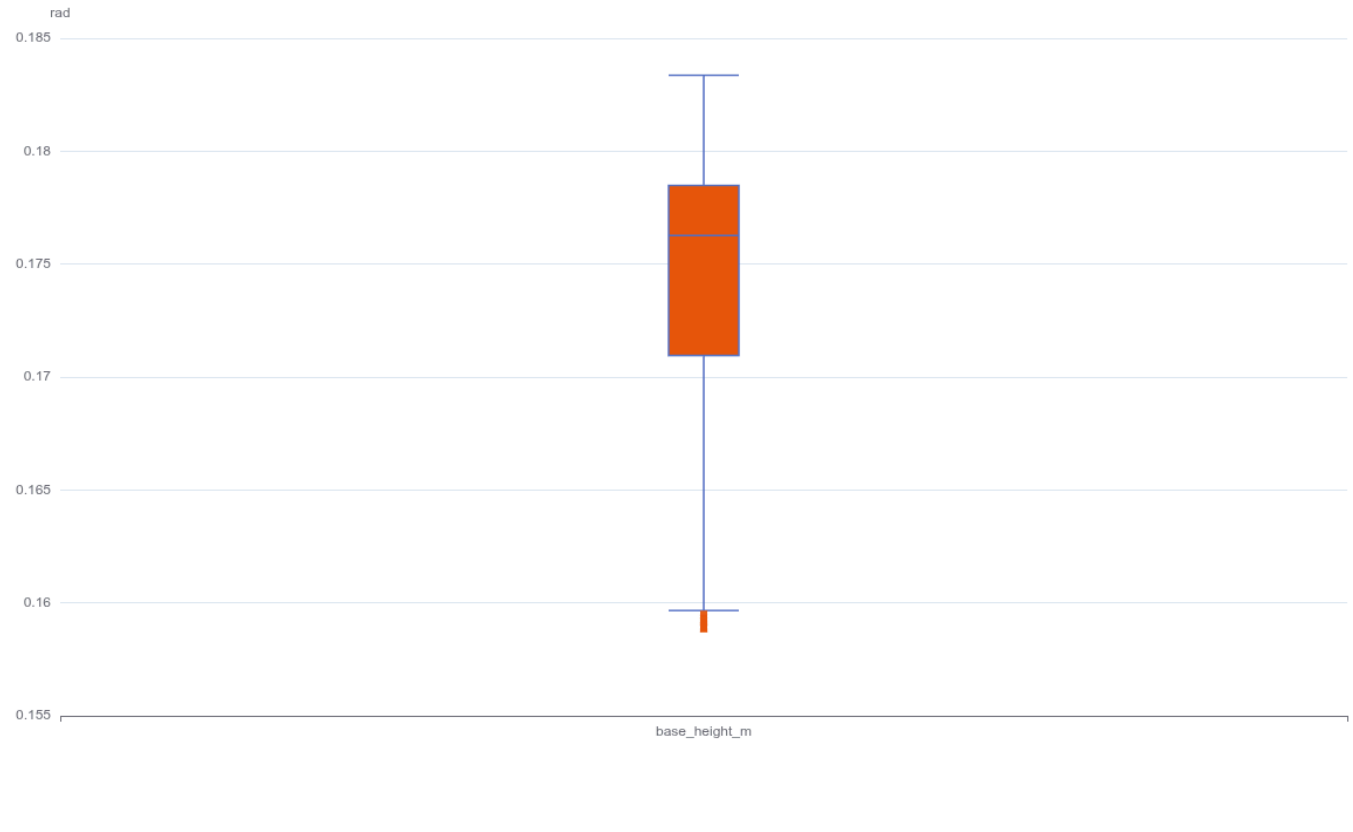

ratio
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

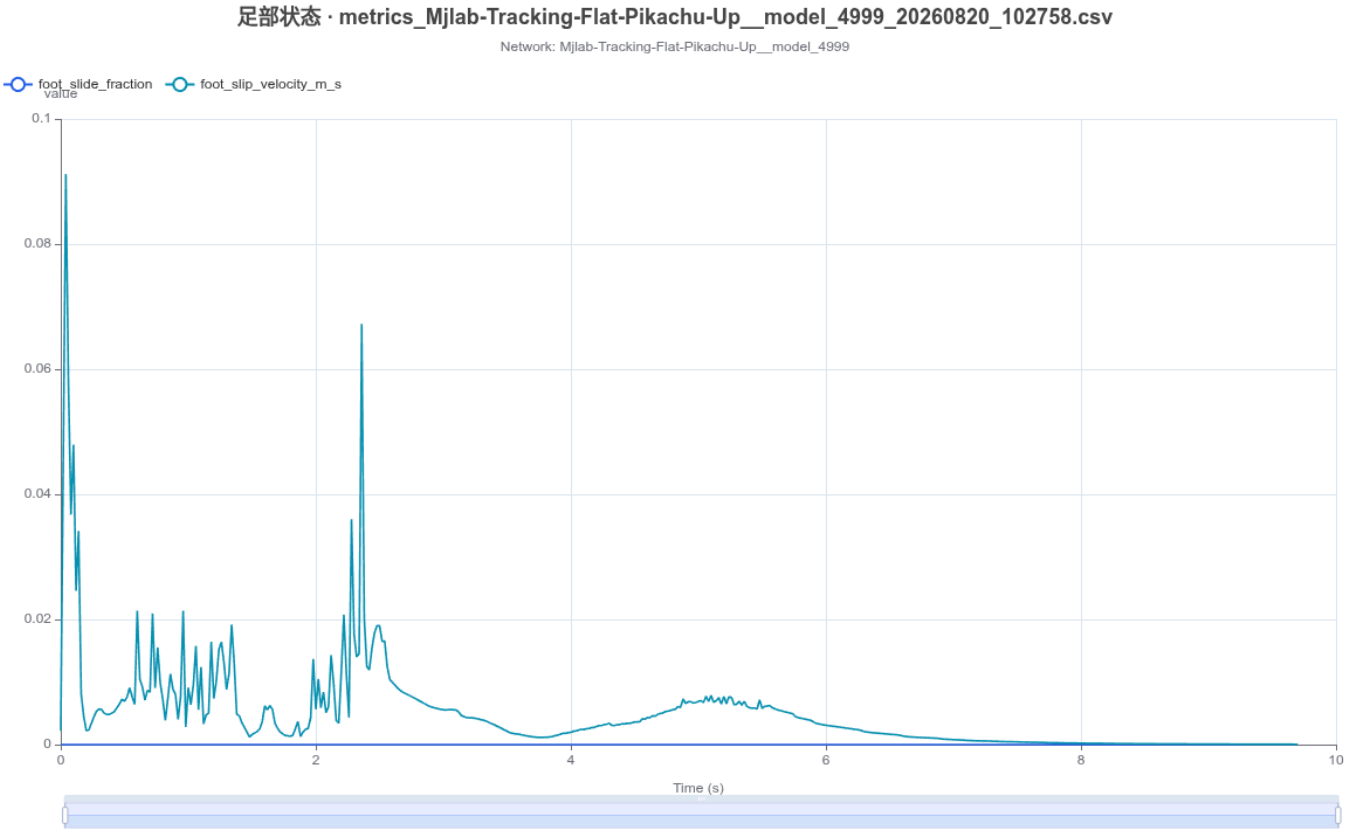
机体状态



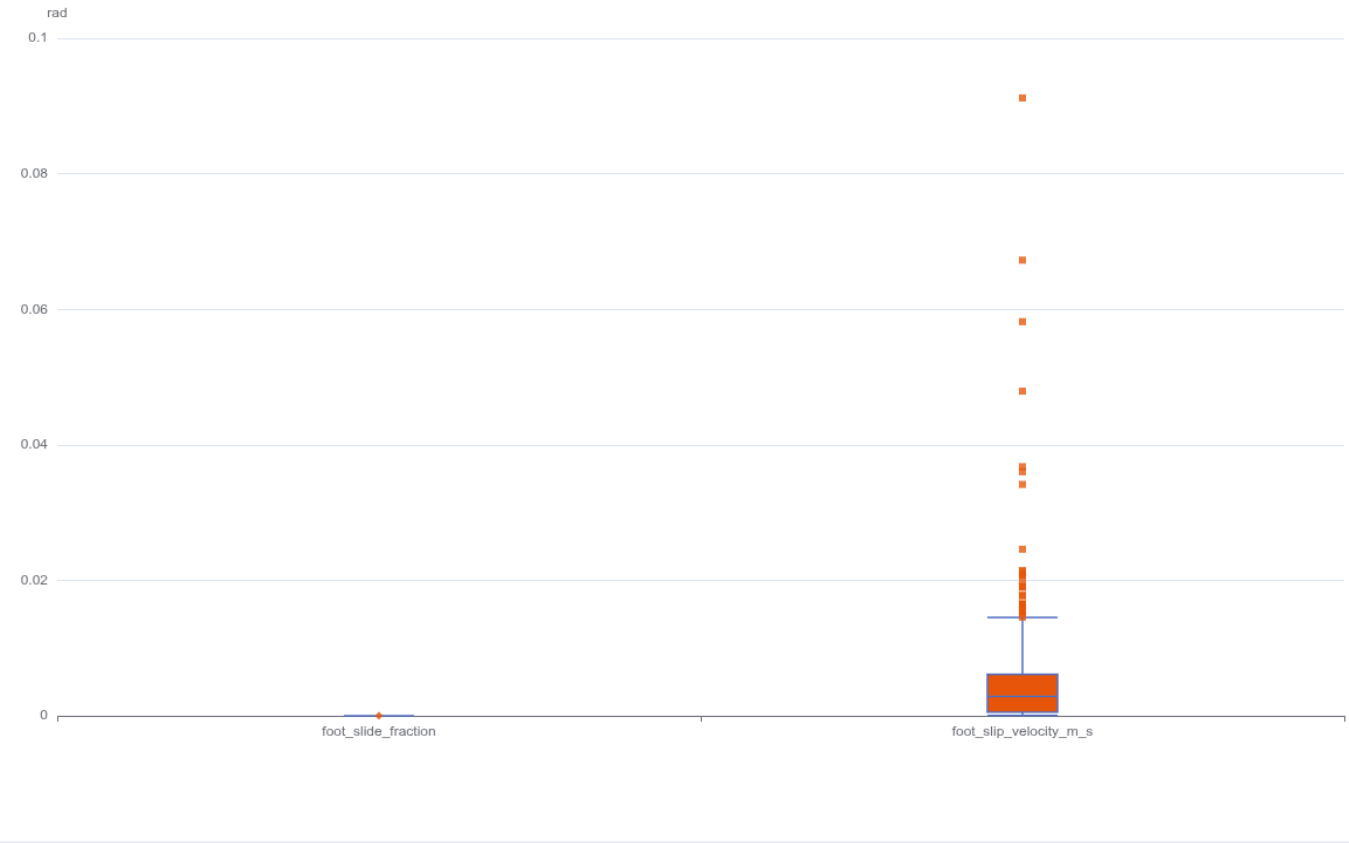
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	455	0.158860	0.183363	0.174253	0.174356	0.183363	0.182688



足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	455	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
foot_slip_velocity_m_s	–	–	455	0.0000475143	0.0911873	0.00481150	0.00919140	0.0911873	0.0155186



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	455	0.00000441819	0.679701	0.00657726	0.0378772	0.679701	0.0222034
actuator_spec_violation	–	–	455	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

